

Article

Investigation of Dead Time Losses in Inverter Switching Leg Operation: GaN FET vs. MOSFET Comparison

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Abstract: This paper investigates the commutation transients of MOSFET and GaN FET devices in motor drive applications during hard-switching and soft-switching commutations at dead time operation. This study compares the switching behaviors of MOSFETs and GaN FETs, focusing on their performance during dead time in inverter legs for voltage source inverters. Experimental tests at various phase current levels reveal distinct switching characteristics and energy dissipation patterns. A validated simulation model estimates the experimental energy exchanged and dissipated during switching transients. The results demonstrate that GaN FETs exhibit lower overall losses at shorter dead times compared to MOSFETs, despite higher reverse conduction voltage drops. The study provides a quantitative framework for selecting optimal dead times to minimize energy losses, enhancing the efficiency of GaN FET-based inverters in low-voltage motor drive applications. Finally, a dead time optimization strategy is proposed and described.

Keywords: motor drive; inverter leg; dead time; switching losses; reverse conduction; GaN FET; MOSFET



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1. Introduction

The performance limitations of silicon-based power devices are increasingly evident, moving the semiconductor industry towards alternative materials like silicon carbide (SiC) and gallium nitride (GaN). GaN, in particular, has gained significant traction due to its superior switching speed capabilities [1,2]. Traditional silicon power MOSFETs have faced challenges in balancing conduction and switching losses, as efforts to reduce on-resistance often result in increased parasitic capacitances, leading to higher switching losses [3].

In Pulse-Width Modulation (PWM) motor drive applications, the adoption of GaN technology offers the potential to achieve higher switching frequencies, which in turn reduces torque ripple and improves the waveform quality of the motor current [4]. In a motor drive powered by a voltage source inverter, the dead time is necessary to avoid cross-conduction [5]. Unfortunately, dead time always causes the waveform distortion phenomenon in a motor drive, and dead time compensation strategies are required [6]. Moreover, voltage source inverters used in these applications require dead time to prevent cross-conduction, introducing waveform distortion. This distortion originates from the inherent delays in switching devices and the characteristics of the devices themselves, such as turn-on and turn-off delays and reverse conduction voltage drop [7].

GaN FETs are particularly attractive in power electronics due to their low on-resistance and ability to operate at very high frequencies [8]. For low-voltage ($V < 100$ V) motor drives, GaN FET-based inverters have demonstrated advantages in reducing the size of passive components and minimizing motor current distortion and torque ripple. On the other hand, an advanced motor insulation layout and a deep investigation of the commutation transient in the inverter leg are required due to the dV/dt increase [9].

Properly setting the dead time is crucial to minimize both reverse conduction [10]. Numerous studies have aimed to optimize dead time settings, exploring solutions like gate driver ICs with adjustable or adaptive dead time capabilities. While some methods, such as programmable dead time settings, lack adaptability in real-world applications [11], others have shown promise but often lack generality or theoretical underpinning [12,13]. Recent research has highlighted the importance of theoretically derived optimal dead time values, which have demonstrated improvements in efficiency [14]. These research studies require a deep knowledge of the considered device behavior, depending on the operative conditions and the technology features [15,16].

This paper investigates the commutation transients of MOSFET and GaN FET devices during dead time for motor drive applications. Experimental tests are conducted in an inverter leg board controlling the phase current. Results reveal different switching behaviors depending on the working conditions. The energy exchanged between the high-side and low-side devices during commutations and the energy losses are estimated through a validated model of the system. The contribution to switching losses during hard-switching and soft-switching commutations and the differences between GaN FET and MOSFET results are distinguished and deeply investigated. Findings aim to provide insights and guidelines for optimizing dead time based on the specific technology for different operating conditions. Furthermore, an optimization strategy for the dead time related to the GaN FET in inverter leg application is presented and described.

2. GaN FET and MOSFET Commutation Transients in Motor Drive Application

The motor drive system used consists of a GaN FET-based inverter powering a 3-phase permanent magnet (PM) motor. Figure 1 shows the system composed of the inverter and the electrical machine. The inverter is composed of three legs, one for each motor phase. The stator phase currents I_a , I_b , I_c are controlled by the high-side Q_{HS} and the low-side Q_{LS} using a Pulse-Width Modulation (PWM). The modulation works at the switching frequency f_{sw} , significantly higher than the AC stator phase current frequency of the motor in order to ensure control stability. A dead time (t_{dt}) is introduced between the devices' commutation in which both driving signals are off-state. This t_{dt} is set by the user to avoid shoot-through in the inverter leg [17]. Nevertheless, the introduction of t_{dt} creates voltage harmonic distortion affecting the phase current waveform [18,19].

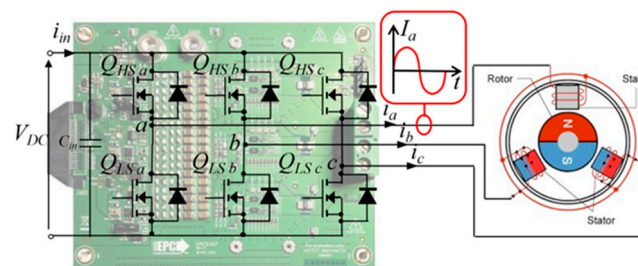


Figure 1. A 2-Level inverter using GaN FETs powering a 3-phase AC permanent magnetic motor.

In AC motor drive systems, each inverter leg operates with a sinusoidal phase current of various amplitudes. These currents are directed either from the inverter leg's switching node to the motor phase or in the reverse direction. Figure 1 shows that a current entering the stator phase is considered positive.

To study commutation transients in switching legs with MOSFET and GaN FET devices, we used two half-bridge experimental board PCBs. These boards only differed in device technology. This setup ensured consistent parasitic effects from the PCB, allowing a fair comparison. Nevertheless, the different packages of GaN FET and MOSFET cannot be removed. However, the choice of the technology leads to the use of the corresponding parasitic elements introduced by the case of the selected device [20,21]. Moreover, tests are

carried out using equal operating conditions for both the GaN FET-based board and the MOSFET-based one.

The GaN FET board featured EPC2065 GaN FET, while the MOSFET one featured Onsemi FDMS2D5N08C. The device features are reported in Table 1.

Table 1. MOSFET Onsemi FDMS2D5N08C and GaN FET EPC2065 features.

Parameter	Symbol	MOSFET	GaN FET
Breakdown voltage	BV_{DSS}	80 V	80 V
Conduction resistance	$R_{DS,on}$	2.7 m Ω	3.6 m Ω
Driving gate voltage	V_g	10 V	5 V

Figure 2a depicts the schematic of the inverter leg. Figure 2b,c show the pictures of the GaN FET board and the MOSFET one, respectively.

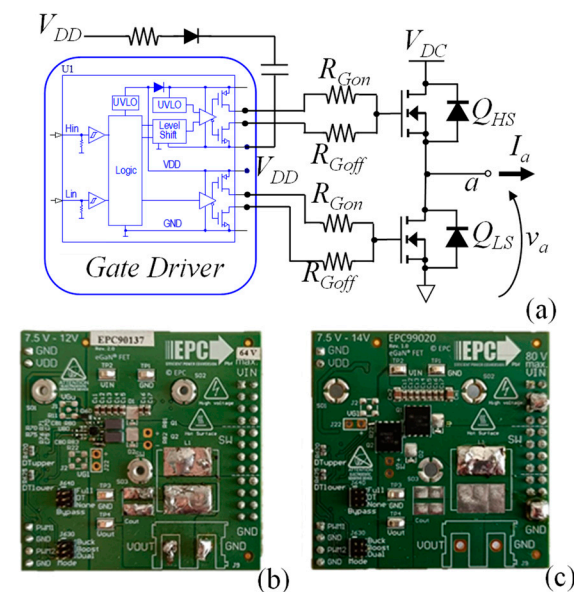


Figure 2. (a) Electric schematic of the half-bridge boards used for testing the devices. (b) GaN FET-based board. (c) MOSFET-based board.

Precise measurement results are particularly challenging to obtain, especially when they aim to distinguish different events that happen in a short time (e.g., during the switching transient of WBG devices). Therefore, a dedicated experimental setup controlling the system variables is required [22].

Testing occurred at an ambient temperature of 25 °C with a DC input voltage of $V_{DC} = 48$ V. A second inverter board is used to control the phase current I_a connecting an LCL filter to the half-bridge switching node (point a of Figure 2a). An STM32H7 microcontroller generated PWM signals and controlled the phase current. The PWM operated at a $f_{sw} = 20$ kHz, switching frequency with a duty cycle of 0.1 to reduce current ripple. This f_{sw} is sufficient to ensure that the switching transient has been completed before a new switching. Despite the fact that WBG devices can operate at a higher switching frequency, the f_{sw} selection does not affect the switching transient's investigation [23]. Additionally, $f_{sw} = 20$ kHz is a reasonable settlement for the MOSFET, which operates at a lower f_{sw} than the GaN FET; t_{dt} duration needs to be chosen long enough to prevent shoot-through and obtain hard-switching for commutations with low phase current [24]. Furthermore, significant distortion effects due to the duration of the dead time need to be avoided [25]. In the experimental test, both GaN FET and MOSFET boards have a dead time of $t_{dt} = 150$ ns, which is a good trade-off between the GaN FET and MOSFET

requirements. During t_{dt} , transitioning from high-side (Q_{HS}) turn-off to low-side (Q_{LS}) turn-on can cause zero-voltage transients at different current levels [26].

Experimental tests carried out with $t_{dt} = 150$ ns reveal that the MOSFET switching leg achieves zero voltage switching (ZVS) [27] for currents $I_a \geq 1.5$ A. The GaN FET achieves ZVS at lower I_a .

The voltage waveforms are measured using a digital scope featuring a bandwidth of 500 MHz, an output resistance of 10 M Ω , and an output capacitance of 10 pF.

The experimental setup of the controlled current-level system is shown in Figure 3. It includes the half-bridge board under test, the power converter regulating I_a , and the STM32H7 microcontroller.

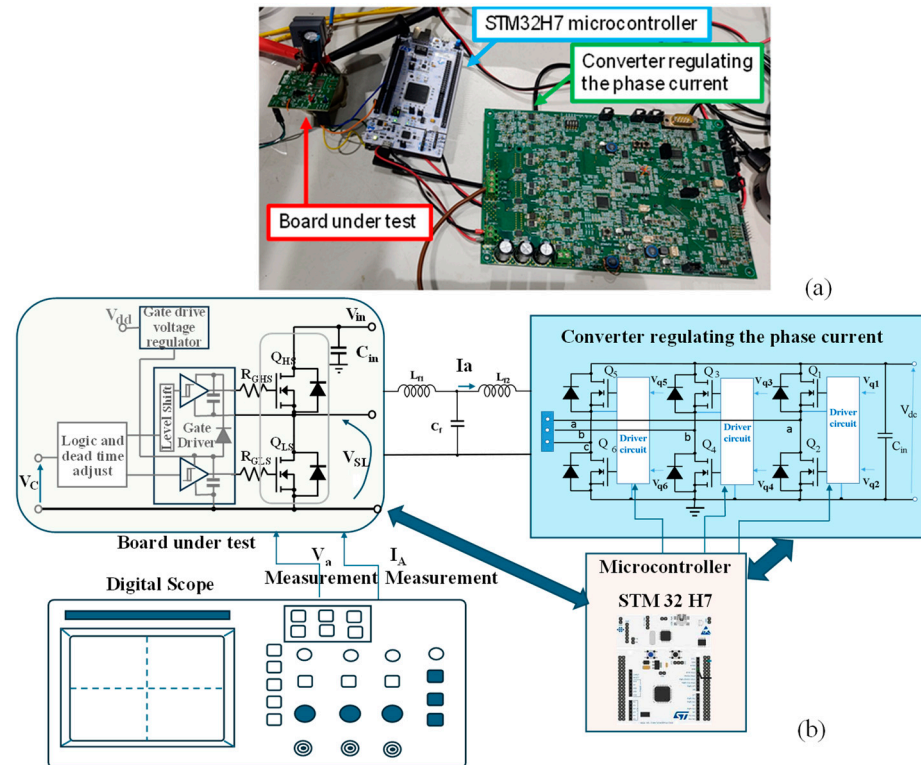


Figure 3. Experimental setup made of the half-bridge board under test (GaN FET and MOSFET), the converter imposing the phase current, and the STM32H7 microcontroller. (a) Picture of the testing bench, (b) block diagram of the experimental setup.

Tests are conducted for positive I_a current values (exiting from the switching node and entering the converter regulating current) at $I_a = 0.5$ A, 1 A, 1.5 A, 2 A, 5 A, 7.5 A, 10 A. Two commutation characteristics for two transitions are analyzed:

- **High-Side Turn-Off, Low-Side Turn-On:** This commutation features a negative voltage slew rate ($dV_a/dt < 0$) as the switching node voltage (V_a) decreases;
- **Low-Side Turn-Off, High-Side Turn-On:** This had a positive voltage slew rate ($dV_a/dt > 0$) as the switching node voltage (V_a) increased to V_{DC} .

The experimental test result achieved in these two commutations for various I_a amplitudes are each reported separately.

2.1. Commutation with $dv_a/dt < 0$ and Positive I_a

Figure 4 shows the voltage waveforms measured on the half-bridge boards using GaN FETs and MOSFETs. Figure 4a displays the switching node voltage V_a , while Figure 4b illustrates the gate-source voltages for the high-side device ($V_{GS,HS}$) and the low-side device ($V_{GS,LS}$). The V_a waveforms correspond to the current amplitudes I_a indicated by the arrows.

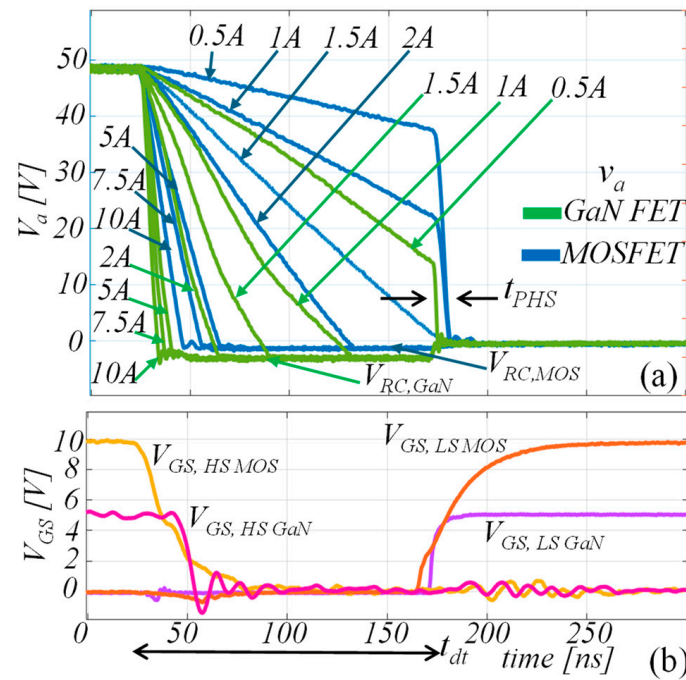


Figure 4. Voltages measured on GaN FET and MOSFET during the Q_{HS} turn-off and Q_{LS} turn-on commutation with $I_a > 0$ A ($t_{dt} = 150$ ns). (a) Switching node voltage V_a . (b) Gate-source voltages V_{GS} . $V_a = 10$ V/div; $V_{GS} = 2$ V/div; timestep = 50 ns/div.

Depending on the t_{dt} length and the I_a , three different switching events can happen for the switching node voltage:

- Zero voltage switching (ZVS);
- Voltage variation and partial hard switching (PHS);
- Voltage fall transient and reverse conduction (RC).

When the high-side switch (Q_{HS}) turns off, V_a starts to fall. The rate of V_a decline is steeper with higher I_a due to the parasitic output capacitances of the devices ($C_{OSS} = C_{GD} + C_{DS}$) and the load [28]. Since C_{OSS} is not constant with voltage, the dynamics of V_a can be described using an equivalent capacitance C_{eq} . This C_{eq} is derived as the average value resulting from the V_a slew rate (dV_a/dt) measured at different I_a amplitudes and considering the time (t_{Vfall}) taken for V_a to fall to 0 V when Q_{HS} turns off. Integrating the constitutive equation of a capacitance ($I = C \cdot dV/dt$), it is possible to calculate C_{eq} as

$$C_{eq} = \frac{V_{DC}}{t_{Vfall} \cdot I_a} \quad (1)$$

ZVS occurs when the low-side switch (Q_{LS}) turns on exactly as V_a reaches 0 V. This is the condition in which $t_{Vfall} = t_{dt}$. As shown in Figure 4, ZVS for MOSFET happens at a phase current amplitude of $I_a = 1.5$ A, while for GaN FET, it occurs between $I_a = 0.5$ A and $I_a = 1$ A. The GaN FET has a lower $C_{OSS} = 750$ pF, compared to the MOSFET's $C_{OSS} = 1800$ pF. The lower C_{OSS} of the GaN FET results in a shorter t_{Vfall} , enabling ZVS at lower I_a compared to the MOSFET.

When I_a is lower than the ZVS threshold, a PHS event follows the V_a transient. In this case, Q_{LS} turns on before V_a has fully dropped to 0 V. After t_{dt} , V_a falls to Q_{LS} 's conduction value within the partial hard switching duration time (t_{PHS}), causing PHS losses. Figure 4 shows MOSFET experiencing PHS at $I_a = 0.5$ A and $I_a = 1$ A, while the GaN FET exhibits PHS only at $I_a = 0.5$ A.

In the cases of higher I_a values than those required for ZVS, V_a drops to 0 V before t_{dt} ends $t_{Vfall} < t_{dt}$. Subsequently, Q_{LS} operates in RC mode until Q_{HS} turns on. The reverse

conduction duration is $t_{RC} = t_{dt} - t_{Vfall}$. V_a is negative at $V_a = -V_{RC}$ due to the activation of the body-diode in the MOSFET or the equivalent diode behavior in the GaN FET. The reverse conduction voltage V_{RC} is higher for GaN FET ($V_{RC} = 1.4$ V) than for MOSFET ($V_{RC} = 0.8$ V), causing higher RC losses in the GaN FET than in the MOSFET. No losses follow t_{dt} as Q_{HS} turns on.

In Figure 5a are highlighted t_{dt} and t_{PHS} in the example of the PHS event achieved with the GaN FET with $I_a = 0.5$ A. Figure 5b indicates the time intervals of t_{Vfall} and t_{RC} relative to the V_a curves achieved with $I_a = 2$ A for the GaN FET.

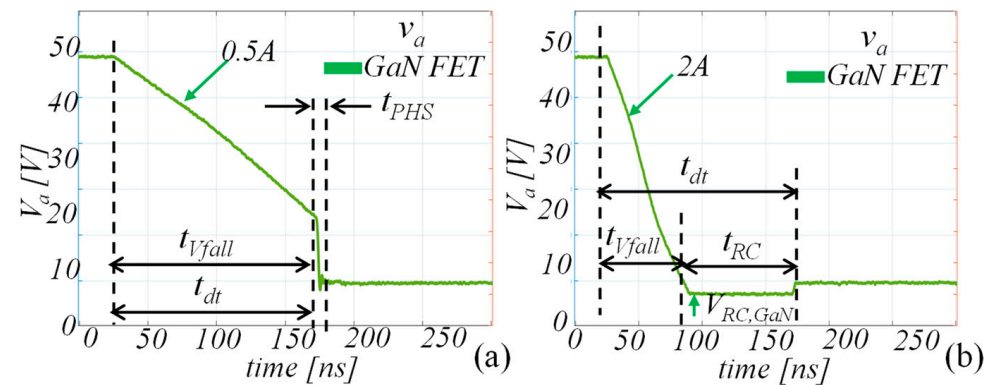


Figure 5. Switching node voltages measure on GaN FET during the Q_{HS} turn-off and Q_{HS} turn-on commutation: (a) PHS event at $I_a = 0.5$ A; (b) RC event at $I_a = 2$ A. $V_a = 10$ V/div; timestep = 50 ns/div.

2.2. Commutation with $dV_a/dt > 0$ and Positive I_a

Figure 6 shows voltage waveforms with $dV_a/dt > 0$ for both the half-bridge board using GaN FETs and the one using MOSFETs at the same current variations as in Figure 4. Figure 6a presents the switching node voltage V_a , while Figure 6b illustrates the gate-source voltages ($V_{GS,LS}$ for the low-side device and $V_{GS,HS}$ for the high-side device).

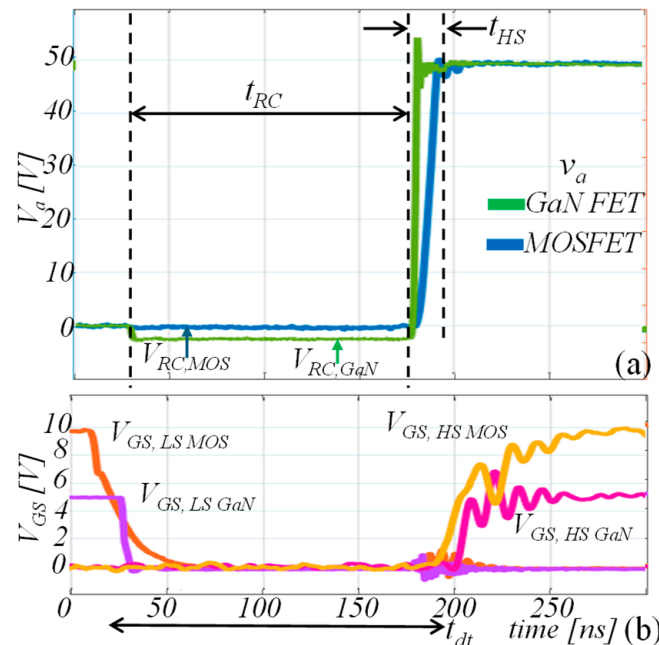


Figure 6. Voltages measured on GaN FET and MOSFET during the Q_{LS} turn-off and Q_{HS} turn-on commutation with $I_a > 0$ A ($t_{dt} = 150$ ns). (a) Switching node voltage V_a . (b) Gate-source voltages at the bottom side. $V_a = 10$ V/div; $V_{GS} = 2$ V/div; timestep = 50 ns/div.

The phase voltage V_a waveforms rise with no differences for all I_a amplitudes used in the tests. The difference in dV_a/dt is determined by the device technology. The parasitic capacitance C_{OSS} of the device affects the voltage rise time t_{HS} . The reverse conduction (RC) phase begins when Q_{LS} turns off and ends when Q_{HS} turns on. During RC, $V_a = -V_{RC}$ due to the body-diode of the MOSFET or the equivalent diode in the GaN FET. A hard switching (HS) event follows the dead time, lasting a few nanoseconds, and is a dissipative process because V_{DS} of Q_{HS} equals $V_a = V_{RC} + V_{DC}$.

2.3. Commutation with $dV_a/dt < 0$ and Negative I_a

The results for negative current are dual to those with positive current. The switching event of $I_a < 0$ A and $dV_a/dt < 0$ is characterized by an RC phenomenon lasting $t_{RC} = t_{dt}$. During RC, V_a reaches the value of $V_a = V_{RC} + V_{DC}$. Consequently, an HS event with a rapid V_a rising happens, and it elapses in lasting a few nanoseconds (t_{HS}). All considerations made for $V_a/dt > 0$ and $I_a > 0$ (Section 2.2) are valid.

Figure 7 illustrates V_a measured on the boards with GaN FETs and MOSFET during the commutation with $dV_a/dt < 0$ when testing with $I_a < 0$ A. Figure 7a depicts the falling V_a waveforms, while Figure 7b shows the gate-source voltages for Q_{LS} and Q_{HS} ($V_{GS,LS}$ for the low-side device and $V_{GS,HS}$ for the high-side device).

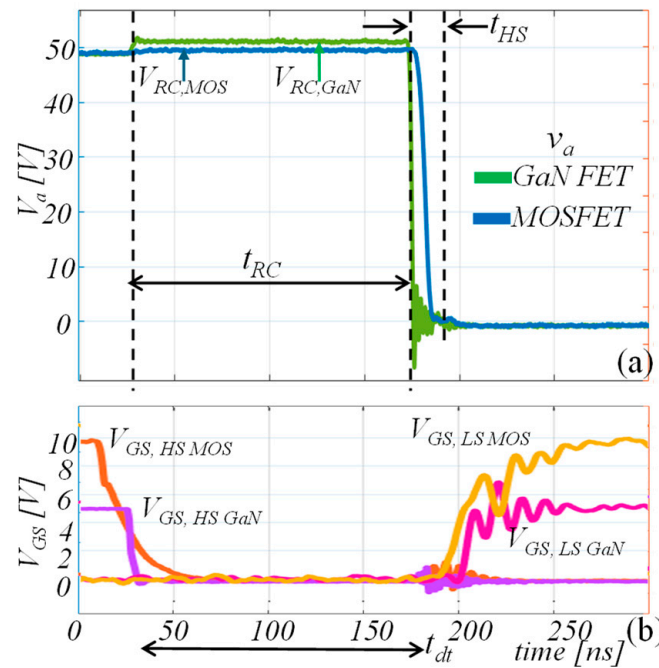


Figure 7. Voltages measured on GaN FET and MOSFET during the Q_{HS} turn-off and Q_{LS} turn-on commutation with $I_a < 0$ A ($t_{dt} = 150$ ns). (a) Switching node voltage V_a . (b) Gate-source voltages. $V_a = 10$ V/div; $V_{GS} = 2$ V/div; timestep = 50 ns/div.

2.4. Commutation with $dV_a/dt > 0$ and Negative I_a

For $dV_a/dt > 0$ and $I_a < 0$ A, the commutation dynamics are influenced by the parasitic capacitance. Depending on the amplitude of I_a , Zero Voltage Switching (ZVS), Partial Hard Switching (PHS), and Reverse Conduction (RC) events can occur. The same considerations discussed for $dV_a/dt < 0$ and $I_a > 0$ A apply here. At higher I_a amplitudes, the voltage rise time (t_{Vrise}) is shorter. Figure 8 shows the voltages measured on GaN FET and MOSFET boards with $I_a = -1.5$ A, -2 A, -5 A, -7.5 A, -10 A during commutation with $dV_a/dt > 0$. Figure 8a illustrates the rising V_a waveforms, while Figure 8b presents the gate-source voltages for Q_{LS} and Q_{HS} ($V_{GS,LS}$ and $V_{GS,HS}$, respectively).

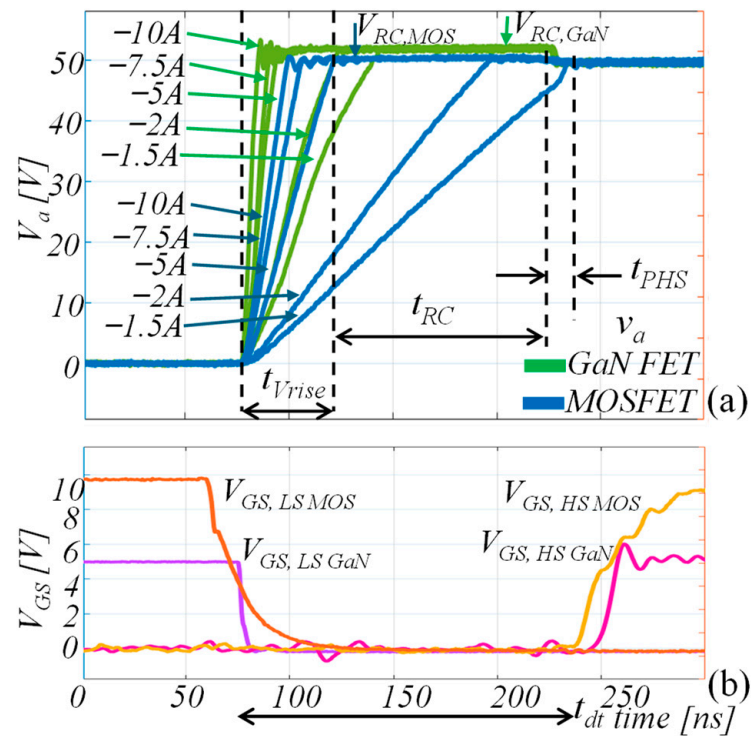


Figure 8. Voltages measured on GaN FET and MOSFET during the Q_{LS} turn-off and Q_{HS} turn-on commutation with $I_a < 0$ A ($t_{dt} = 150$ ns). (a) Switching node voltage V_a . (b) Gate-source voltages V_{GS} . $V_a = 10$ V/div; $V_{GS} = 2$ V/div; timestep = 50 ns/div.

3. GaN FET vs. MOSFET Commutation Energy Evaluation

The board used for experimental tests does not incorporate current sensing to measure the current of transistors in a half-bridge configuration. This is intentionally designed to prevent any impact on the switching board's performance. Nevertheless, measuring transistor current is crucial for evaluating power trends and energy during commutations. To achieve the current waveform of devices, LTSpice simulations are used. The simulation models for MOSFET and GaN FET are sourced from the manufacturer's official websites.

The simulated electrical circuit replicates the experimental setup (see Figure 2a) and maintains the same operating conditions: $V_{DC} = 48$ V; $f_{sw} = 20$ kHz and duty-cycle 0.1. The half-bridge circuit model is validated by ensuring it produces waveforms consistent with those obtained in the experimental tests, as depicted in Figures 4 and 6 [29]. The phase currents exiting the switching node (positive I_a) in the simulations are $I_a = 0.5$ A, 1 A, 1.5 A, 2 A, 5 A, 7.5 A, 10 A, emulating the experimental conditions.

Simulations are performed twice with dead times $t_{dt} = 20$ ns and $t_{dt} = 150$ ns. These t_{dt} values are typical for the respective devices (20 ns for GaN FETs and 150 ns for MOSFETs) and are relevant for motor drive applications. The energy values computed from the simulations pertain to the low-side device (Q_{LS}).

Figures 9 and 10 show the Q_{LS} waveforms of the current I_{LS} , the phase voltage V_a , and the power P_{LS} during the commutation with $dV_a/dt < 0$, load current of $I_a = 2$ A, and a dead time of $t_{dt} = 20$ ns and with $t_{dt} = 150$ ns, respectively. Figures 9a and 10a refer to the GaN FET, while Figures 9b and 10b refer to the MOSFET.

Comparing the GaN FET waveforms of Figures 9a and 10a with the MOSFET ones in Figures 9b and 10b shows that the V_a and current variations last longer in the MOSFET than in the GaN FET. As a result, during the V_a fall the GaN FET power P_{GAN} has a peak comparable with those of the MOSFET P_{MOS} , but P_{GAN} lasts shorter. In the case of soft switching following the V_a fall in Figure 9a,b, GaN FET features a higher reverse conduction voltage drop (V_{RC}). The current flowing through the device operating in reverse conduction is I_a and it is equal for both the GaN FET or the MOSFET. Therefore, the higher V_{RC} of

the GaN FET leads to higher reverse conduction losses than the MOSFET [30]. Due to the higher voltage drop of the GaN transistor during reverse conduction operation, the device in the third quadrant must work with reduced timing to optimize performance and losses. In the hard-switching event with $I_a = 2$ A and $t_{dt} = 20$ ns shown in Figure 10a,b, the overall GaN FET hard switching losses are much lower than the MOSFET ones.

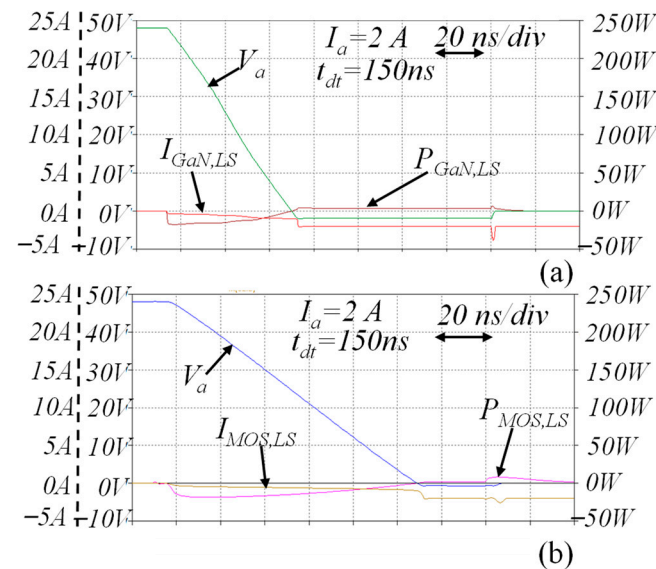


Figure 9. Waveforms of the commutation with $I_a = 2$ A, $t_{dt} = 150$ ns, and $dV_a/dt < 0$. (a) GaN FET $I_{GaN} = 10$ A/div; $V_a = 10$ V/div; power $P_{GaN} = 25$ W/div. (b) MOSFET $I_{MOS} = 10$ A/div; $V_a = 10$ V/div; power $P_{MOS} = 25$ W/div. timestep = 20 ns/div.

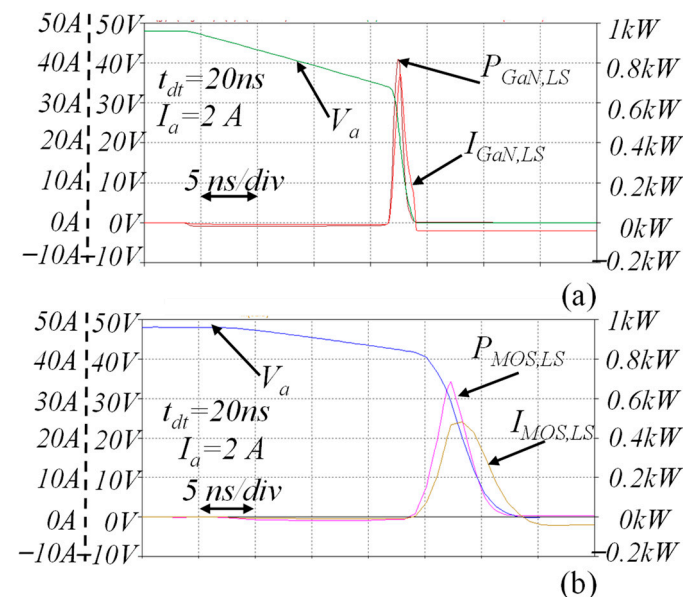


Figure 10. Waveforms of the commutation with $I_a = 2$ A, $t_{dt} = 20$ ns, and $dV_a/dt < 0$. (a) GaN FET $I_{GaN} = 10$ A/div; $V_a = 10$ V/div; power $P_{GaN} = 200$ W/div. (b) MOSFET $I_{MOS} = 10$ A/div; $V_a = 10$ V/div; power $P_{MOS} = 200$ W/div. timestep = 5 ns/div.

During the commutation exhibiting $dV_a/dt < 0$ (Q_{HS} turning off and Q_{LS} turning on) with a positive I_a , it is possible to distinguish between the I_a amplitudes that result in ZVS

or RC and those that cause PHS. The energy of Q_{LS} during the voltage variation, E_{VV} , is determined by the equation:

$$E_{VV} = \int_0^{t_{Vfall}} |V_a \cdot I_{LS}| \cdot dt \quad (2)$$

where I_{LS} is the current through Q_{LS} and t_{Vfall} is the time taken for the switching node variation V_a . The time t_{Vfall} varies according to the V_a falling slew rate. If V_a drops to 0 V before the end of t_{dt} , RC conditions appear. ZVS condition appears when t_{Vfall} equals the maximum value of t_{dt} if V_a reaches 0 V at the end of t_{dt} . The PHS conditions occur if the theoretical voltage fall time t_{Vfall} exceeds t_{dt} . When t_{Vfall} is followed by RC conditions lasting t_{RC} , the energy losses during t_{dt} are denoted as E_{RC} and calculated as

$$E_{RC} = \int_{t_{Vfall}}^{t_{Vfall}+t_{RC}} |V_{RC} \cdot I_{LS}| \cdot dt \quad (3)$$

where t_{RC} is the RC time interval when V_a is negative and equal to $V_a = -V_{RC}$ for the GaN FET or MOSFET. The upper integral limit at $t = t_{Vfall} + t_{RC}$ is close to t_{dt} and includes turn-on delays of Q_{LS} and the driving circuit's propagation delay uncertainty.

If V_a does not drop to 0 V within t_{dt} , the voltage variation during $t_{Vfall} = t_{dt}$ is followed by PHS. The energy losses due to the PHS E_{PHS} are calculated as

$$E_{PHS} = \int_{t_{dt}}^{t_{dt}+t_{PHS}} |V_a \cdot I_{LS}| \cdot dt \quad (4)$$

where t_{PHS} is the duration of the PHS phenomenon, starting at the end of t_{dt} and ending as $I_{LS} = 0$ A.

In the ZVS condition, $E_{RC} = 0$ J and $E_{PHS} = 0$ J. E_{VV} is the only energy involved in the switching event.

Figures 11 and 12 illustrate the energies involved during the Q_{HS} turn-off and Q_{LS} turn-on with positive I_a from 0.5 A to 10 A. Figure 11 refers to results with $t_{dt} = 20$ ns, while Figure 12 refers to those achieved with $t_{dt} = 150$ ns. In particular, Figures 11a and 12a depict the energy E_{VV} as a function of I_a . On the other hand, Figures 11b and 12b show the energy losses due to RC (E_{RC}) and PHS (E_{PHS}) as a function of I_a . ZVS is marked with a dashed line, while PHS and RC are marked with dotted lines. MOSFET curves are blue, and GaN FET curves are green.

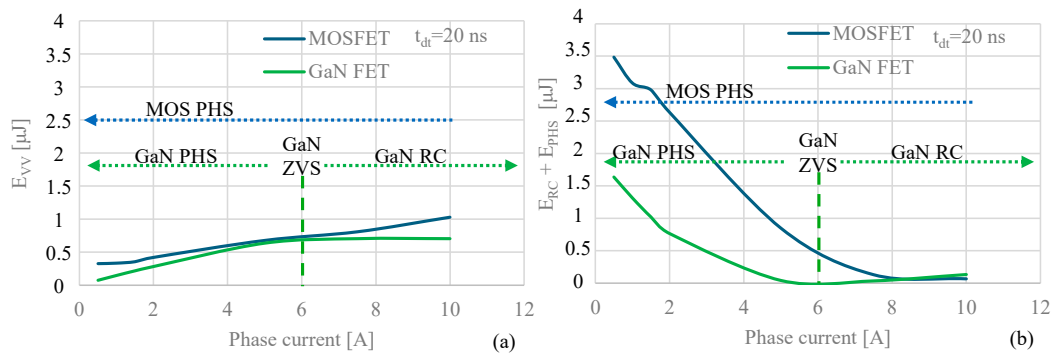


Figure 11. Energies of Q_{LS} (MOSFET in blue and GaN FET in green) versus phase current during the commutation with $I_a > 0$ A and $dV_a/dt < 0$ using $t_{dt} = 20$ ns. (a) Energy of voltage variation; (b) PHS and RC energies. ZVS event is highlighted with a dashed line. PHS and RC are highlighted with dotted lines. $E = 0.5 \mu\text{J}/\text{div}$; $I_a = 2 \text{ A}/\text{div}$.

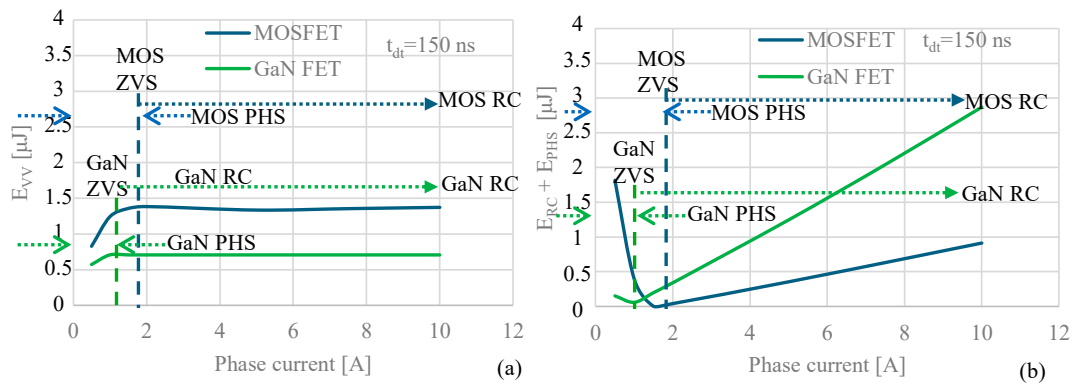


Figure 12. Energies of Q_{LS} (MOSFET in blue and GaN FET in green) versus phase current during the commutation with $I_a > 0$ A and $dV_a/dt < 0$ using $t_{dt} = 150$ ns. (a) Energy of voltage variation; (b) PHS and RC energies. ZVS event is highlighted with a dashed line. PHS and RC are highlighted with dotted lines. $E = 0.5 \mu\text{J}/\text{div}$; $I_a = 2 \text{ A}/\text{div}$.

GaN FET achieves ZVS at $I_a = 6$ A with $t_{dt} = 20$ ns and at $I_a = 0.8$ A with $t_{dt} = 150$ ns. Differently, the MOSFET achieves ZVS at $I_a = 1.5$ A with $t_{dt} = 150$ ns, while it does not achieve ZVS with $t_{dt} = 20$ ns for currents up to $I_a = 10$ A. RC losses occur for I_a higher than the one causing ZVS, while PHS occurs for I_a lower than the ZVS.

Commutation with $dV_a/dt < 0$ starts with the initial conditions of $C_{OSS,LS}$ of Q_{LS} charged to V_{DC} and $C_{OSS,HS}$ of Q_{HS} discharged to nearly 0 V. Immediately after turning off Q_{HS} , $C_{OSS,LS}$ of Q_{LS} discharges to 0 V and $C_{OSS,HS}$ of Q_{HS} charges to V_{DC} . During this, both devices are in the off-state, but the variation of V_a causes an exchange of energy E_{VV} between them. The amount of E_{VV} exchanged varies depending on the switching event:

- ZVS or RC: $C_{OSS,LS}$ of Q_{LS} discharges of E_{VV} until V_a stabilizes at 0 V, while $C_{OSS,HS}$ of Q_{HS} charges of E_{VV} up to V_{DC} . E_{VV} is maximum (E_{VVmax}) since $C_{OSS,LS}$ of Q_{LS} fully discharges using the charging energy of $C_{OSS,HS}$ of Q_{HS} . E_{VVmax} is calculated as

$$E_{VVmax} = \frac{1}{2} \cdot C_{eq} \cdot V_{DC}^2 \quad (5)$$

and looking at Figures 11a and 12a, $E_{VVmax} \approx 0.75 \mu\text{J}$ for GaN FET and $E_{VVmax} \approx 1.5 \mu\text{J}$ for MOSFET.

- PHS: $C_{OSS,LS}$ of Q_{LS} does not discharge completely because after t_{dt} the phase voltage is not null, $V_a > 0$ V. $C_{OSS,HS}$ of Q_{HS} charges of $E_{VV} < E_{VVmax}$, and $C_{OSS,LS}$ of Q_{LS} does not fully charge to $V_a = V_{DC}$. GaN FET's smaller C_{OSS} results in a steeper V_a fall and a shorter t_{Vfall} for exchanging E_{VV} . As a result, MOSFET features a greater quantity of E_{PHS} losses than the GaN FET when using the same t_{dt} .

ZVS condition features zero energy losses $E_{RC} + E_{PHS} = 0$ J. Only E_{VV} is involved due to the energy exchange between the switching leg devices' output capacitances ($C_{OSS,HS}$ and $C_{OSS,LS}$). Differently, PHS and RC are dissipative phenomena following t_{Vfall} (Figures 11b and 12b).

Considering the same I_a , PHS energy losses E_{PHS} is lower with long t_{dt} because Q_{LS} turns on with a lower $V_{DS} = V_a > 0$ V (closer to ZVS). Additionally, elapsed t_{dt} , GaN FET's smaller C_{OSS} results in a steeper V_a fall than MOSFET, leading to lower E_{PHS} for GaN FET. On the other hand, MOSFET features a lower voltage drop V_{RC} and corresponding losses E_{RC} than the GaN FET. E_{RC} increase proportionally with t_{dt} and I_a whatever the technology is considered.

Figures 13 and 14 show the GaN FET and MOSFET waveforms of the Q_{LS} current I_{LS} , the phase voltage V_a and the device power P_{LS} during the commutation with $dV_a/dt > 0$. Additionally, Figures 13a and 14a show the GaN FET power P_{GaN} , while Figures 13b and 14b depict the MOSFET reverse recovery current and the body-diode

power, respectively. In particular, Figure 13 refers to the case with $I_a = 2$ A and Figure 14 to $I_a = 7.5$ A.

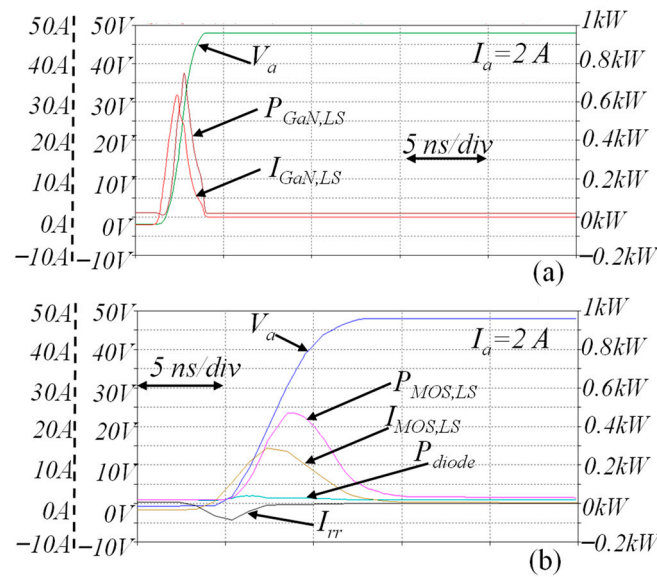


Figure 13. Waveforms of the commutation with $I_a = 2$ A and $dV_a/dt > 0$. (a) GaN FET $I_{GaN} = 5$ A/div; $V_a = 5$ V/div; power $P_{GaN} = 100$ W/div. (b) MOSFET $I_{MOS} = 5$ A/div; $V_a = 5$ V/div; power $P_{MOS} = 100$ W/div; reverse recovery current $I_{rr} = 5$ A/div; body-diode power $P_{diode} = 100$ W/div. timestep = 5 ns/div.

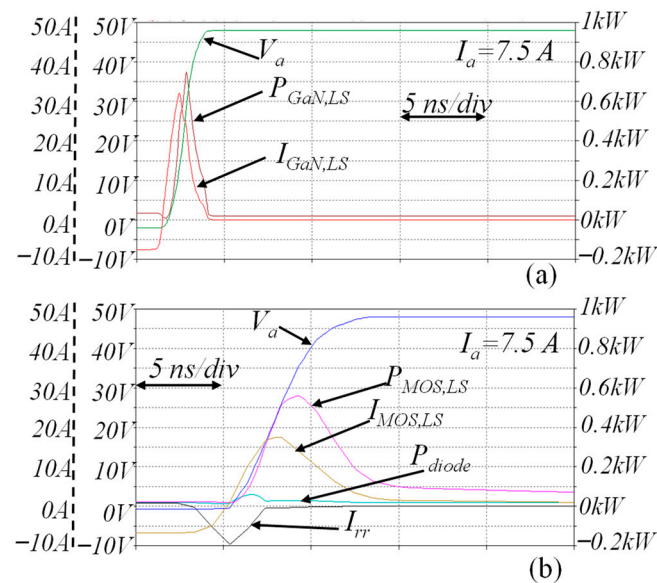


Figure 14. Waveforms of the commutation with $I_a = 7.5$ A and $dV_a/dt > 0$. (a) GaN FET $I_{GaN} = 5$ A/div; $V_a = 5$ V/div; power $P_{GaN} = 100$ W/div. (b) MOSFET $I_{MOS} = 5$ A/div; $V_a = 5$ V/div; power $P_{MOS} = 100$ W/div; reverse recovery current $I_{rr} = 5$ A/div; body-diode power $P_{diode} = 100$ W/div. timestep = 5 ns/div.

During commutation with $dV_a/dt > 0$ (Q_{LS} turn off and Q_{HS} turn on), the transient event starts with the $C_{OSS,LS}$ of Q_{LS} being discharged, and $C_{OSS,HS}$ of Q_{HS} being charged to V_{DC} . With positive I_a , Q_{LS} works in reverse conduction for all t_{dt} . At the end of t_{dt} , Q_{HS} turns on with a drain-source voltage of $V_a = V_{DC}$, causing hard switching (HS). The device Q_{LS} has a current peak and an almost instantaneous $C_{OSS,LS}$ charge.

GaN FET waveforms in Figures 13a and 14a exhibit negligible differences with the I_a level. Furthermore, GaN FET shows a faster dynamic than the MOSFET one of

Figures 13b and 14b because of the low GaN FET C_{OSS} . Additionally, MOSFET features a reverse recovery current I_{rr} due to the body-diode which causes an increase of P_{diode} in switching losses P_{MOS} . A higher I_a leads to a higher I_{rr} peak and higher P_{diode} , as shown comparing Figures 13b and 14b. GaN FET does not feature reverse recovery current and corresponding losses. The I_{GaN} peak is due only to the charging C_{OSS} .

The reverse conduction energy E_{RC} achieved by Q_{LS} both for GaN FET and MOSFET is illustrated in Figures 15a and 16a as a function of I_a . Figures 15b and 16b depict the hard switching energy losses (E_{HS}) as a function of I_a . Figure 15 refers to the case with $t_{dt} = 20$ ns, while Figure 16 refers to the one with $t_{dt} = 150$ ns. GaN FET curves are depicted in green, while MOSFET curves are shown in blue.

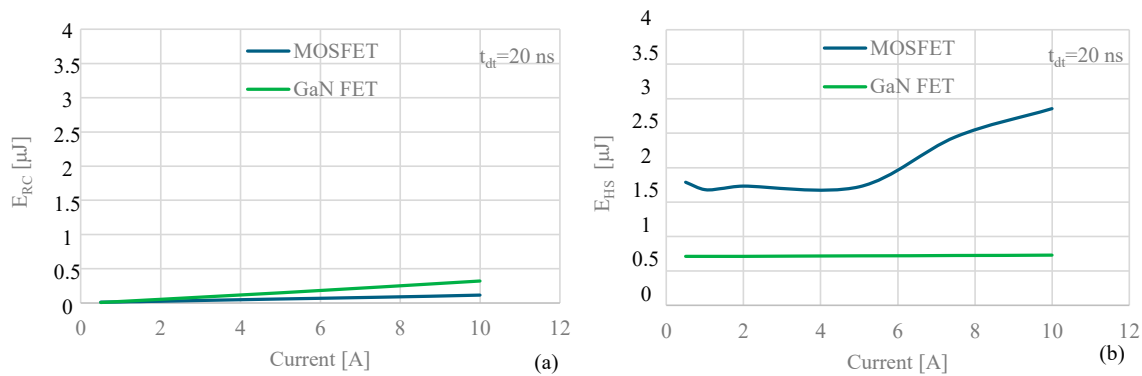


Figure 15. Energies of Q_{LS} (MOSFET in blue and GaN FET in green) versus phase current I_a during the commutation with $I_a > 0$ A and $dV_a/dt < 0$ using $t_{dt} = 20$ ns. (a) Energy dissipation for reverse conduction E_{RC} ; (b) Energy dissipation for hard switching E_{HS} . $E = 0.5 \mu J/\text{div}$; $I_a = 2 \text{ A}/\text{div}$.

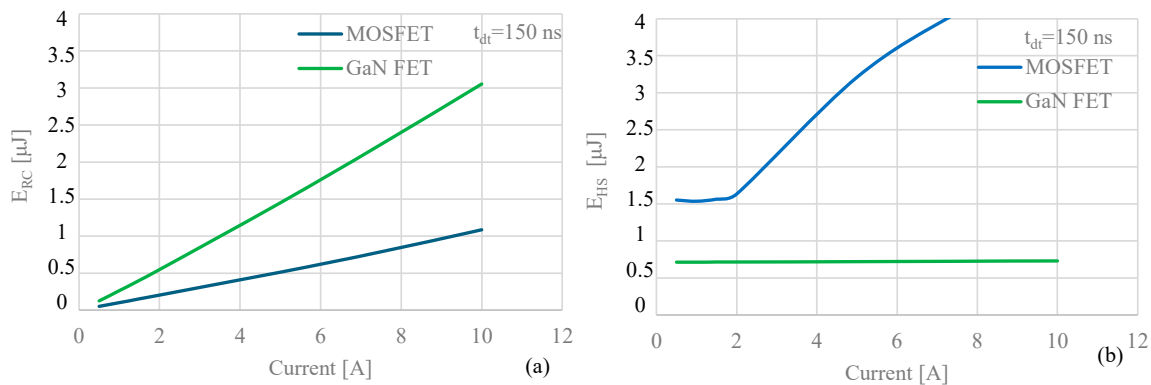


Figure 16. Energies of Q_{LS} (MOSFET in blue and GaN FET in green) versus phase current I_a during the commutation with $I_a > 0$ A and $dV_a/dt < 0$ using $t_{dt} = 150$ ns. (a) Energy dissipation for reverse conduction E_{RC} ; (b) Energy dissipation for hard switching E_{HS} . $E = 0.5 \mu J/\text{div}$; $I_a = 2 \text{ A}/\text{div}$.

The comparison of the GaN FET and MOSFET E_{RC} curves in Figures 15a and 16a demonstrates that GaN FET losses are higher than MOSFET during t_{RC} due to higher V_{RC} [30]. Despite this, E_{RC} losses are significantly low (up to 0.3 μJ for GaN FET and 0.1 μJ for MOSFET) considering $t_{dt} = 20$ ns. E_{RC} increases linearly with t_{dt} and the level of I_a .

Figures 15b and 16b show that HS energy losses E_{HS} of GaN FET are significantly lower than the MOSFET ones. GaN FET features a $E_{HS} = 0.75 \mu J$ regardless of t_{dt} and I_a . Differently, MOSFET, and E_{HS} has a minimum $E_{HSmin} = 1.5 \mu J$ for low I_a levels which persist according to t_{dt} duration: $I_a < 6$ A for $t_{dt} = 20$ ns and $I_a < 2$ A for $t_{dt} = 150$ ns.

E_{HSmin} corresponds to the energy required to charge the $C_{OSS,LS}$ of Q_{LS} to V_{DC} (no energy comes from $C_{OSS,HS}$ of Q_{HS}) and it is calculated as

$$E_{HSmin} = \frac{1}{2} \cdot C_{eq} \cdot V_{DC}^2 \quad (6)$$

where C_{eq} is the constant equivalent value of C_{OSS} (which is a non-linear parameter with the voltage). E_{HSmin} is not affected by t_{dt} and closely matches the maximum energy capacity of C_{OSS} (E_{VVmax}), as arises comparing in Figures 12a and 16b.

When I_a level is higher, E_{HS} of the MOSFET grows with the I_a amplitude, while the GaN FET one remains constant. The MOSFET E_{HS} increase is due to the additional reverse recovery charge Q_{rr} in the MOSFET's P-N junction, which grows with longer t_{dt} and higher reverse recovery current I_{rr} [31,32]. Conversely, the GaN FET maintains a constant $E_{HS} = E_{HSmin}$ due to the absence of a P-N junction, thereby featuring $Q_{rr} = 0$ nC [33]. In general, E_{HS} significantly exceeds E_{RC} , particularly at $t_{dt} = 20$ ns. When considering both energy dissipation components, the GaN FET exhibits reduced energy losses during the switching transient, especially with a shorter dead time ($t_{dt} = 20$ ns).

4. Dead Time Reduction Strategy

A reduction switching losses strategy can be developed acting on the dead time length. In particular, it is aimed to set the dead time differently for the switching events of $dV_a/dt < 0$ (Q_{HS} turn off and Q_{LS} turn on) and $V_a/dt > 0$ (Q_{LS} turn off and Q_{HS} turn on), according to the load current sign:

1. $I_a > 0$ A
 - a. $dV_a/dt > 0$: set $t_{dt} = t_{dt,min}$
 - b. $dV_a/dt < 0$: set $t_{dt} = t_{dt,opt}$
2. $I_a < 0$ A
 - c. $dV_a/dt > 0$: set $t_{dt} = t_{dt,opt}$
 - d. $dV_a/dt < 0$: set $t_{dt} = t_{dt,min}$

Where $t_{dt,min}$ is the minimum dead time which ensures a safe switching event according to the gate propagation delay uncertainty; $t_{dt,opt}$ is the optimum dead time minimizing the reverse conduction duration. Furthermore, in motor drive applications, it may be useful to maintain $t_{dt,opt}$ between a maximum value $t_{dt,max}$ and the minimum $t_{dt,min}$. The dead time $t_{dt,max}$ is chosen according to the maximum admissible hard-switching losses and avoids increasing the total harmonic distortion (THD) in the phase motor current [33,34].

4.1. Operation with Constant Minimum Dead Time Conditions

When turning on GaN FET HS with a $I_a > 0$ A, case a, (see Figure 6) and GaN FET LS with $I_a < 0$ A, case d, (see Figure 7), reverse conduction phenomenon persists for all the dead time duration, regardless of the I_a amplitude. Reverse conduction is minimized by simply setting the minimum t_{dt} duration ($t_{dt,min}$). The $t_{dt} = t_{dt,min}$ condition can be enabled after having monitored the I_a sign and applied only for the turn-on transient of the corresponding GaN FET: HS turns on when $I_a > 0$ A; LS turns on when $I_a < 0$ A.

4.2. Operation with Reduced Dead Time Conditions

The control strategy for turning on GaN FET LS when $I_a > 0$ A, case b, (see Figure 4), or GaN FET HS when $I_a < 0$ A, case c, (see Figure 8) requires a preliminary estimation of the optimum dead time $t_{dt,opt}$. A comparator is used to monitor the drain-source voltage V_{DS} of the GaN FET to turn on, as depicted in the circuit schematic of Figure 17a. A constant reference voltage V_{refON} is set as a threshold. The comparator compares V_{DS} with a threshold reference value V_{refON} . The dead time control block in Figure 17a generates the driving signal V_q when V_{DS} falls below V_{refON} , as shown in Figure 17b. Additionally, a maximum dead time value $t_{dt,max}$ in the dead time control block is considered [35].

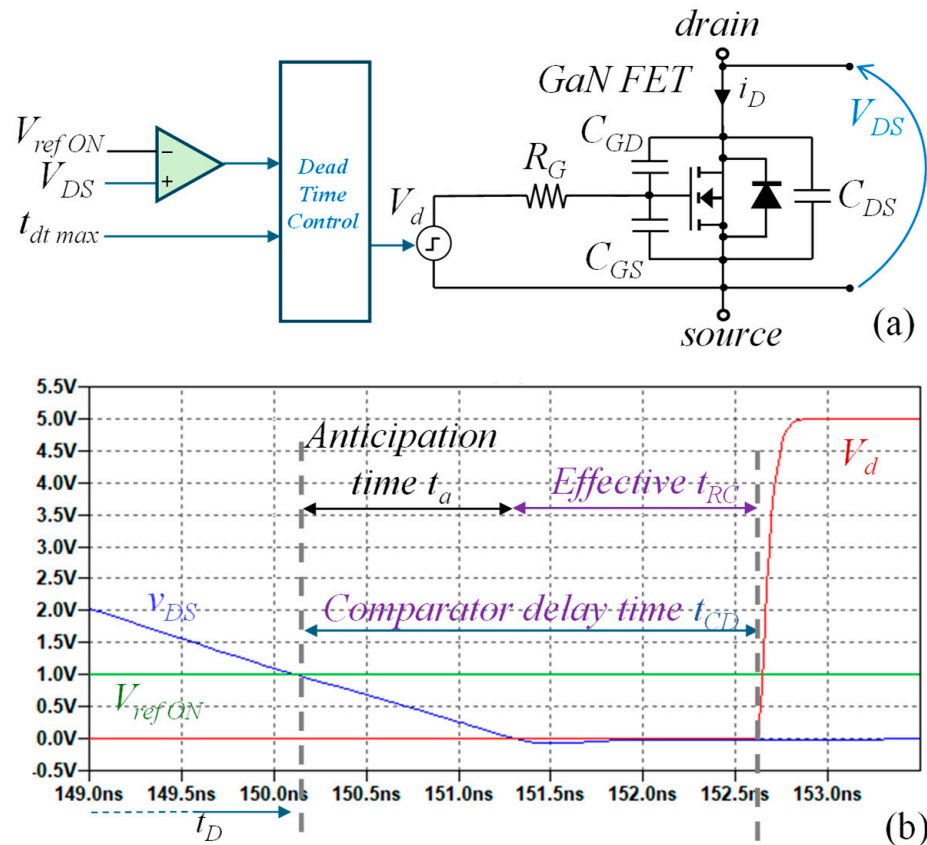


Figure 17. (a) v_{DS} comparison and $t_{dt,max}$ entering the dead time control strategy for dead time reduction. (b) Choice of V_{refON} for v_{DS} comparison. $V = 0.5 \text{ V/div}$. $\text{timestep} = 5 \text{ ns/div}$.

If $V_{DS} < V_{refON}$ is not triggered within the maximum dead time $t_{dt,max}$, the turn-on signal is generated regardless of the comparator output (the dead time is fixed and equal to $t_{dt,max}$).

In Figure 17b, t_d represents the turn-on driving time which corresponds to the time when the condition $V_{DS} = V_{refON}$ is triggered. The threshold V_{refON} is chosen according to the highest I_a amplitude that does not cause reverse conduction within $t_{dt,max}$. This condition is equivalent to ZVS with $t_{dt} = t_{dt,max}$. V_q generation can be anticipated by the comparator delay time t_{cd} . V_{refON} is set equal to the value of V_{DS} when $t_D = t_{dt,max} - t_{cd}$.

This approach ensures that the turn-on signal is sent only when the voltage variation has been completed, thereby minimizing reverse conduction. The commutation behavior is thus adapted to the current amplitude, differing from constant dead time methods.

4.3. Validation in Motor Drive Setup

For instance, the presented dead time reduction strategy is developed in a GaN FET-based inverter board supplying a BLDC motor (nominal voltage of 36 V, nominal power of 250 W, and 26 pole pairs). A maximum dead time of $t_{dt,max} = 100 \text{ ns}$ and a minimum of $t_{dt,min} = 20 \text{ ns}$ are selected. At the time $t_{dt,max} = 100 \text{ ns}$ the I_a current in ZVS is 1.2 A. For the phase current below 1.2 A, the PHS condition appears. This $t_{d,mx}$ value choice is a trade-off between the quality of the sinusoidal output current and reduced energy losses during the partial hard switching conditions (E_{PHS} contribute).

The V_a waveforms measured for $I_a = 1.2 \text{ A}$, 1.6 A, 2.5 A, 4 A, 6 A utilizing the dead time reduction strategy are shown in Figure 18a for commutations with $dV_a/dt < 0$ and in Figure 18b for commutations with $dV_a/dt > 0$.

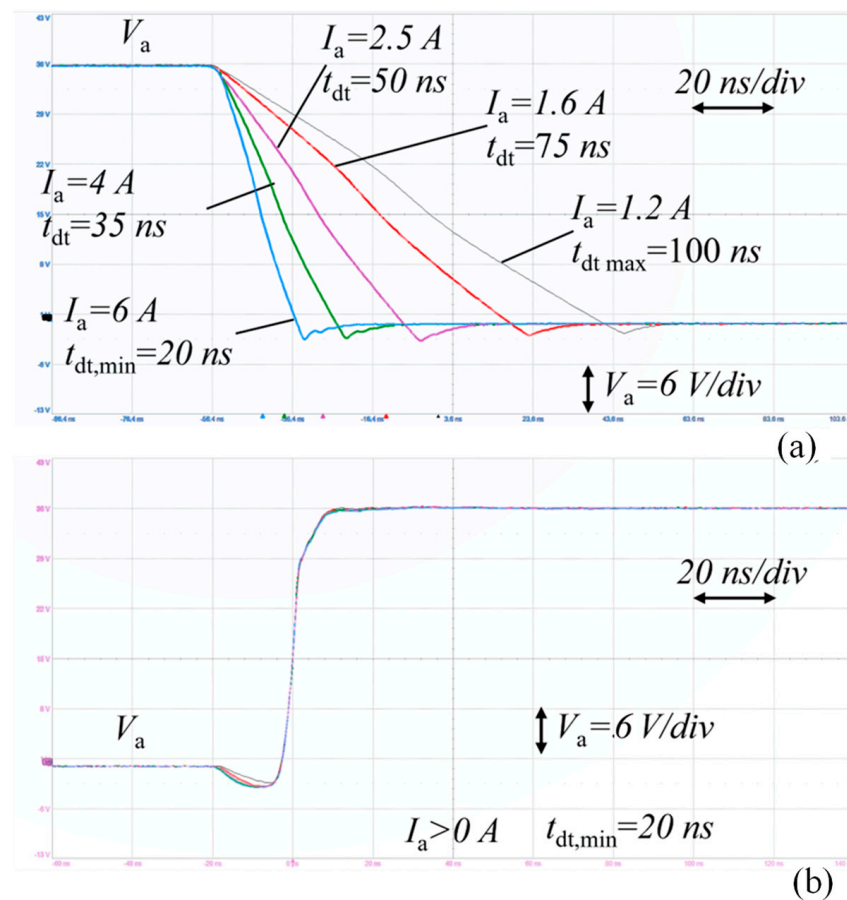


Figure 18. V_a waveforms measured for $I_a = 1.2 \text{ A}$, 1.6 A , 2.5 A , 4 A , 6 A utilizing the dead time reduction strategy. (a) $dV_a/dt < 0$; (b) $dV_a/dt > 0$; $V_a = 5 \text{ V/div}$. timestep = 20 ns/div .

The V_a voltage fall in Figure 18a shows that with higher I_a it is possible to minimize the reverse conduction phenomenon shortening the dead time. Moreover, for $I_a < 1.2 \text{ A}$ the dead time is maximum $t_{dt, \max} = 100 \text{ ns}$, while for $I_a > 6 \text{ A}$ the dead time is minimum $t_{dt, \min} = 20 \text{ ns}$. On the other hand, in Figure 18b, when V_a rises with $I_a > 0 \text{ A}$, the dead time is minimum $t_{dt, \min} = 20 \text{ ns}$ independently from the I_a amplitude.

5. Conclusions

This paper presents a comprehensive investigation into the commutation transients of MOSFET and GaN FET devices in motor drive applications, with a focus on hard-switching and soft-switching commutations. Through experimental tests and validated simulations, the study reveals distinct differences in switching behaviors and energy dissipation patterns between MOSFETs and GaN FETs. The key findings highlight that GaN FETs exhibit significantly lower overall losses at shorter dead times compared to MOSFETs, despite a higher reverse conduction voltage drop. Additionally, the lower output parasitic capacitance of GaN FETs contributes to faster commutations and reduced energy losses.

These insights provide a quantitative framework for optimizing dead time duration to minimize energy losses in GaN FET-based low-voltage inverters for motor drive applications. Furthermore, a strategy to optimize the dead time choice for the different operative conditions in the inverter leg is presented and described. The proposed strategy for dynamically adjusting dead time based on load conditions shows potential for further reducing commutation losses and enhancing inverter efficiency. Future research will focus on developing algorithms to adapt dead time duration in real-time, thereby optimizing performance and reducing energy consumption in motor drive systems.

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Conflicts of Interest: Author Marco Palma was employed by the Efficient Power Conversion Corporation. The remaining authors declare that the research was conducted in the absence of any commercial or financial relationship that could be construed as a potential conflict of interest.

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